# Dosage Calculation & IV Drip Rates

Formulas, worked examples and common traps for nursing medication math and NCLEX review

MindShelf

Contents


1 Core methods....1
2 Unit conversions....1
3 Weight-based & safe dose....1
4 IV infusion rates....2
5 Body surface area (Mosteller)....2
6 Reconstitution & strength....2
7 Titration & drips....2
8 Rounding & syringes....3
9 Safe notation....3
10 Medication rights & orders....3


*Drugs A–F and their numbers: exercise values, not dosing guidance.*

## 1 Core methods

**Three methods, one answer**

**Desired over have:** $V = D/H \times Q$. $D$ dose ordered, $H$ dose on hand, $Q$ volume (or tablets) holding $H$. **Ratio–proportion:** $H : Q = D : V$, then cross-multiply. **Dimensional analysis:** chain the order through conversion factors until only the wanted unit remains.

**EXAMPLE Injection, three ways (Drug A)**

Drug A ordered 28 mg; vial 32 mg per 1 mL.
D/H × Q: 28 ÷ 32 × 1 = 0.875 ⇒ **0.88 mL**
Ratio 32 : 1 = 28 : $V$ and DA agree. Use a 1 mL syringe (0.01 mL marks).

**TRAP**

Divide only once units match: g against mg is off by a factor of 1,000.

## 2 Unit conversions

**The ×1,000 ladder**

g → mg → mcg, L → mL: × 1,000 per step down, ÷ per step up. lb → kg: × 0.45359237 (exact; courses ÷ 2.2). in → cm: × 2.54 (exact).

| US measure | NIST (exact, rounded) | Label & course |
|---|---|---|
| 1 tsp | ≈ 4.93 mL | 5 mL |
| 1 tbsp | ≈ 14.79 mL | 15 mL |
| 1 fl oz | ≈ 29.57 mL | 30 mL |
| 1 cup | ≈ 236.6 mL | 240 mL |

**EXAMPLE Child's weight**

44 lb × 0.45359237 ≈ 19.958 ⇒ **20 kg** (recorded to 0.1 kg); course shortcut 44 ÷ 2.2 = 20 kg.

**CLINICAL NOTE**

Oral liquids: dose in mL with an oral syringe, never a kitchen spoon.

**TRAP**

Pounds typed as kg: the dose is about 2.2 times too high.

## 3 Weight-based & safe dose

dose = mg/kg × kg; **per day** ÷ doses per day = **per dose**. Safe if low end × kg ≤ ordered daily total ≤ high end × kg.

**EXAMPLE Safe-range check (Drug B)**

17–33 mg/kg/day; 20 kg; ordered 165 mg every 8 h. Min 17 × 20 = 340 mg/day; max 33 × 20 = 660 mg/day. Ordered 165 × 24/8 = 495 mg/day: safe. Outside the range: hold, call the prescriber.


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**TRAP**
mg/kg/day is a 24 h total: compare per day or per dose, like with like.

## 4 IV infusion rates

**Pump:** rate (mL/h) = $V$ (mL) ÷ $T$ (h). **Gravity:** drip (gtt/min) = $V$ × DF ÷ ($T$ × 60). DF (gtt/mL) is printed on the tubing: macrodrip 10, 15 or 20; microdrip 60. **60-drop shortcut:** with microdrip, gtt/min = mL/h. **End time** = start + $V$ ÷ rate.

**EXAMPLE Same order, pump vs gravity**
1,000 mL over 8 h. Pump: 1,000 ÷ 8 = 125 mL/h.
Gravity, DF 20: 1,000 × 20 ÷ (8 × 60) ≈ 41.67 ⇒ **42 gtt/min**.

**EXAMPLE End time and microdrip**
750 mL at 100 mL/h from 09:00: 750 ÷ 100 = 7.5 h (7 h 30 min) → ends 16:30. Microdrip: 100 × 60 ÷ 60 = 100 gtt/min.

**TRAP**
gtt/min needs minutes: hours alone give a rate 60 times too high.

## 5 Body surface area (Mosteller)

$$BSA\ (m^2) = \sqrt{\text{height (cm)} \times \text{weight (kg)} \div 3{,}600}$$

**EXAMPLE BSA-based dose (Drug C)**
150 cm, 54 kg: $\sqrt{150 \times 54 \div 3{,}600} = \sqrt{2.25}$ = 1.5 m². Ordered 92 mg/m²: 1.5 × 92 = 138 mg.

**TRAP**
Mosteller's 3,600 assumes cm and kg: convert in and lb first.

## 6 Reconstitution & strength

concentration = drug ÷ **final** volume on the label. % (w/v) = g per 100 mL: 0.9% = 0.9 × 1,000 ÷ 100 = 9 mg/mL. Ratio 1:1,000 = 1 g in 1,000 mL = 1 mg/mL.

**EXAMPLE Powder vial (Drug D, fictional label)**
1.4 g; label: add 6.2 mL to obtain 7 mL. 1.4 × 1,000 ÷ 7 = 200 mg/mL. Order 320 mg: 320 ÷ 200 = 1.6 mL (3 mL syringe, 0.1 mL marks).

**TRAP**
Powder adds volume: use the final 7 mL, not the 6.2 mL of diluent.

## 7 Titration & drips

$C$ (mcg/mL) = mg in bag × 1,000 ÷ mL in bag. mL/h = mcg/kg/min × kg × 60 ÷ $C$. Reverse: mcg/kg/min = mL/h × $C$ ÷ (kg × 60). Units: mL/h = (units/h) ÷ (units/mL).

**EXAMPLE mcg/kg/min → mL/h (Drug E)**
360 mg in 300 mL: 360 × 1,000 ÷ 300 = 1,200 mcg/mL. 4.5 mcg/kg/min, 76.4 kg (as recorded): 4.5 × 76.4 × 60 ÷ 1,200 = 17.19 ⇒ **17.2 mL/h**.

**EXAMPLE units/kg/h → mL/h (Drug F)**
30,000 units in 400 mL = 75 units/mL. 9.5 units/kg/h × 64 kg = 608 units/h; 608 ÷ 75 ≈ 8.107 ⇒ **8.1 mL/h**.

**TRAP**
Bag in mg, dose in mcg: × 1,000 first; per-minute doses need × 60.

**CLINICAL NOTE**
ISMP high-alert (IV heparin, insulin): independent double check per policy.

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## 8 Rounding & syringes

**Round once, at the end**

- Weight: record to 0.1 kg, use as recorded. Keep concentrations and intermediates unrounded.
- gtt/min: whole drops. mL/h: pump precision (0.1 mL/h here; some pumps: whole mL/h).
- Syringe: to its marks, 0.01 mL (1 mL size) or 0.1 mL (3 mL size).

**TRAP**
Rounding a concentration or intermediate early shifts the final rate.

## 9 Safe notation

| Never write | Risk | Write |
|---|---|---|
| 1.0 mg | Read as 10 mg (trailing zero) | 1 mg |
| .5 mg | Read as 5 mg (no leading zero) | 0.5 mg |
| U or u | Read as 0 or 4 | unit(s) |
| IU | Read as IV or 10 | international unit(s) |
| QD / QOD | Daily and every other day confused | daily / every other day |
| µg | Read as mg | mcg |
| cc | Read as u (units) | mL |
| MS, MSO4, MgSO4 | Morphine and magnesium confused | full drug name |
| 100000 units | Zeros miscounted | 100,000 units |

**TRAP**
A trailing zero or a missing leading zero makes a tenfold dose error.

## 10 Medication rights & orders

**Rights** (patient, drug, dose, route, time) need system safeguards, not just a checklist. **Complete order:** patient name and ID, date/time, drug, dose with unit, route, frequency, weight if weight-based, signature.

**CLINICAL NOTE**
Unclear or incomplete order: clarify with the prescriber before giving.

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